MAX9295A RevID=8 (HS78A-0C) ERRATA SHEET

Data sheet 19-100421; RevE; 12/19 and earlier Errata

RevID=8 (per reading of register 0x0E) Engineering Prototypes

The errata listed below describe situations where MAX9295A components perform differently than expected or differently than described in the data sheet.

This errata sheet only applies to MAX9295A components.

1. **UART - The most significant 8 bits of the 16-bit register address cannot change during an auto-incremented access**

   Description:
   In UART multi-byte read/write access, the auto-incremented address cannot cross 0x**FF. For example, if the first address accessed is 0x10FE, then the maximum length of the access cannot exceed 2 bytes. Any larger access will cause the access to fail in this example.

   Root cause:
   When the 8 LSB bits are 1's, the design prevents the address from incrementing.

   Workaround:
   None

   Resolution:
   No silicon fix currently planned.

2. **Watermark error not latched**

   Description:
   The watermark error register is cleared when read. However, if the error persists while the register is being read, the error is not latched again.

   Root cause:
   The sticky error register is cleared to 0 on read, instead of setting to the current error value.

   Workaround:
   Read the WM_ERR register after reading the interrupt, to check if the error persists.

   Resolution:
   No silicon fix currently planned.

3. **GMSL1 Spread Spectrum amplitude must be programmed only when video input is active**

   Description:

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The GMSL1 link has spread spectrum capability with default spread of 0.5%. If user desires to change the config_spread_bit_ratio from the default value, it needs to be programmed in particular sequence in order to function as intended.

Workaround:
The bitfield, config_spread_bit_ratio, should be programmed only after the video input starts, and also each time the video input stops and restarts.

Resolution:
No silicon fix currently planned.

4. GPIO pin glitch

Description:
If VDDIO is brought up before VDD18, there is a potential for a signal glitch on any GPIO pin. Also, if VDDIO is brought up while the power manager is off, the GPIO could end up in an unknown state. However, once all the power supplies are up, the part will be in the default state.

Workaround:
Ensure that VDD18 is brought up at the same time or before VDDIO.

Resolution:
Planned to be fixed in the next revision.

5. Functional problems with GPIO TX cascade mode

Description:
By default, multiple GPIO toggles (transitions) can be sent in a series in cascading GPIO toggles within the same GMSL packet. Disabling this behavior, forcing each toggle to be sent in a separate GMSL packet, can cause unreliable GPIO transmission across the GMSL link. Due to this issue, the GPIO TX cascade mode should be left enabled.

Workaround:
Leave the bit-field in it's default state: GPIO_TX_CASC=1.

Resolution:
Planned to be fixed in a future revision.

6. Watermark register decode information incorrect

Description:
In register WM_0, decode information for field WM_MODE is incorrect. Correct information is:

- 0b000 = Default generator mode - cycle through all 4 watermarks in video stream (default)
- 0b001 = Error generator mode - cycle through only 2 watermarks to replicate a frozen frame error condition.
- 0b010 - 0b111 = Reserved

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Workaround:
Ensure that correct settings are used.

Resolution:
Data sheet ECN is planned.

7. UART permanent bypass mode can disable access to main control channel if link lock is lost

Description:
If UART permanent bypass mode is enabled, access to main control channel may become disabled, if Link LOCK is lost, and even after link LOCK is restored. The device will need to be power cycled to re-enable access to the main control channel.

Workaround:
Do not use permanent bypass mode in register BYPASS_TO.

Resolution:
No silicon fix currently planned.

8. Termination resistor for unused GMSL2 output in coax mode should be 49.9Ω 1%.

Description:
In the “Typical GMSL1/2 Link-Application Circuit for Coax Cable” figure, the termination resistor for the inverting SIO pin is indicated as 51Ω. Maxim recommends a 49.9Ω 1% resistor be used.

Workaround:
For future builds use 49.9Ω, 1% resistors in termination resistor locations for coax applications.

Resolution:
Data sheet ECN is planned.

9. GPIO values received are not retained in sleep mode

Description:
When GPIO values are received from the other side of the link, they are not retained when going to sleep mode. This is because the GPIO logic is incorrectly put in reset before the received values can be retained in the retention memory.

Workaround:
There are several options. The first is to not use sleep mode. The second is to ensure the GPIO is set to a static value before entering sleep mode. This can be done by disabling GPIO receiving (GPIO_RX_EN = 0) and set the GPIO output value in the GPIO_OUT bit.

Resolution:
Data sheet ECN is planned.

10. I2C/UART timeout status bits are not latched

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Description:
The “I2C_TIMED_OUT” status bit, at register address 0x47, was intended to indicate when an I2C transaction has hung and has exceeded the timeout threshold. However, the status indicator is not latched and will automatically clear immediately following assertion. The immediate clearing of the status makes this status indicator essentially unusable.

Workaround:
With a robust I2C interface, a timeout event should generally not occur. As an alternate method of validating proper I2C transactions, the master I2C device should always observe and respond appropriately to the presence or absence of I2C acknowledge (ACK) indicators.

Resolution:
Data sheet ECN is planned.

11. SPI clock SCLK pulse width may not always match as programmed

Description:
The SPI SCLK pulse width will increase by 3.3ns from the programmed value on the final cycle prior to slave select de-assertion. This will not adversely affect the SPI operation.

Workaround:
None needed. Ignore any variation in SCLK pulse width, as it will not adversely affect the SPI operation.

Resolution:
No silicon fix planned.

12. GMSL Lock can be lost due to high jitter in a temperature window

Description:
An internal voltage regulator requires a register write to increase its output voltage to avoid excessive jitter on the GMSL link. This jitter can lead to loss of GMSL lock.

Workaround:
Perform the register write below.
Register: 0x302
Write Data: 0x10
Bit(s): PFDDIV_RSHORT[2:0]
Purpose: Increase regulator voltage to the clock system to ensure robust operation

Resolution
No silicon fix is planned.

13. SPI master maximum clock rate reduced to 25Mhz

Description:
Due to final test limitations the maximum supported rate for the SPI has been reduced to 25MHz.

Workaround:

Single Link Mode:
Reduce maximum SPI clock output rate to 25MHz by setting the following registers:

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SPIM_SCK_LO_CLKS >= 4
SPIM_SCK_HI_CLKS >= 4

Dual Link Mode or Splitter Mode:
Reduce maximum SPI clock output rate to 25MHz by setting the following registers:
SPIM_SCK_LO_CLKS >= 8
SPIM_SCK_HI_CLKS >= 8

Resolution:
Datasheet ECN planned.

14. ADC has systematic gain error when used with input divider in /2, /3 or /4 Mode and when monitoring internal supplies

Description:
The input dividers in front of the on-board ADC have small systematic deviations from the ideal values /2, /3 and /4 that are used to monitor voltages higher than the reference voltage for the ADC. The deviations from the ideal divide ratios 2, 3 and 4 depend on selected divide ratio and selected ADC input.

Workaround:
When monitoring external voltages, use the following divide ratios in calculating ADC output code

	Divide Ratio		
ADC Input	**/2 Mode**	**/3 Mode**	**/4 Mode**
ADC0	2.011	3.010	4.011
ADC1	2.046	3.079	4.118
ADC2	2.046	3.079	4.118

When monitoring internal supply voltage rails, use the following divide ratios in calculating ADC output code

Monitored Voltage Rail	Divide Ratio
VDDIO/4	4.013
VDD18/2	2.016
CAP_VDD/2	2.009

Resolution:
Data sheet ECN is planned.

15. ADC input Multiplexer is incorrectly documented in data sheet

Description:
Input 0xA of the ADC input multiplexer (register 0x501, bits [7..4]) should be described as CAP_VDD, not VDD in data sheet figure 32. The ADC input multiplexer is incorrectly documented in data sheet
Workaround:
Use register definition is update in table below.

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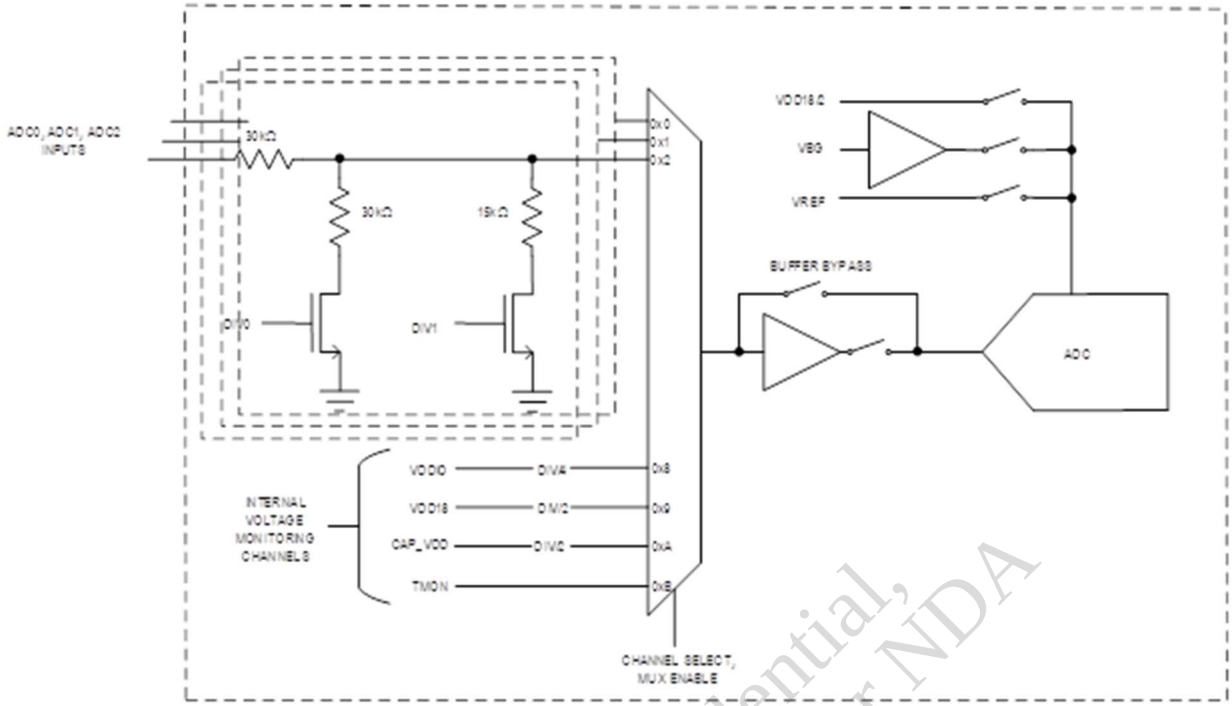

Resolution:
Data sheet ECN is planned.

16. ADC registers incorrectly documented in data sheet

Description:
The following table shows registers that need updated descriptions

Register	Bitfield	Bits	Description	Decode
ADC_CTRL_1 (0x501)	adc_chsel	7:4	0: ADC0 1: ADC1 2: ADC2 8: VDDIO/4 9: VDD18/2 10: CAP_VDD/2 11: TEMP MON	0x0: ADC0 0x1: ADC1 0x2: ADC2 0x8: VDDIO/4 0x9: VDD18/2 0xA: CAP_VDD/2 0xB: TEMP MON
ADC_LIMIT0_3 (0x517)	ch_sel0			
ADC_LIMIT1_3 (0x51B)	ch_sel1			
ADC_LIMIT2_3 (0x51F)	ch_sel2			
ADC_LIMIT3_3 (0x523)	ch_sel3			
ADC_LIMIT4_3 (0x527)	ch_sel4			
ADC_LIMIT5_3 (0x52B)	ch_sel5			
ADC_LIMIT6_3 (0x52F)	ch_sel6			
ADC_LIMIT7_3 (0x533)	ch_sel7			
ADC_CTRL_2 (0x502)	adc_tst_din	5	Reserved	Leave at default value for correct ADC operation
	adc_tst_dout	4	Reserved	Leave at default value for correct ADC operation

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	adc_div	3:2	ADC<2:0> internal divider	0: Divide by 1 1: Divide by 2 2: Divide by 3 3: Divide by 4
ADC_DATA0 (0x508)	adc_data_l	7:0	Lower byte of 10bit ADC converted sample data output	
ADC_RR_CTRL1 (0x535)	No bits should be customer visible			

Workaround:
N/A

Resolution:
Data sheet ECN is planned.

17. Built-in Temperature Monitor (TMON) temperature error may exceed typical error stated in data sheet

Description:
The junction-temperature monitor (TMON) is described in the data sheet as having "typical error over the specified junction temperature range on the order of +/-10 degC". This integrated temperature monitor is not intended for applications requiring precise temperature measurement, and has only been characterized at the high end of the temperature range. A more accurate statement is "Typical error of the part is +/-15°C at 105°C." This error is only an estimate. It is not a guaranteed specification.

Workaround:
None. Do not use the TMON for precision junction temperature sensing.

Resolution:
Datasheet ECN planned.

Revision History

REVISION NUMBER	REVISION DATE	DESCRIPTION	PAGES CHANGED
A	February 28, 2020	Initial release	—
B	October 13, 2020	Added "GMSL Lock can be lost due to high jitter in a temperature window" Added "SPI master maximum clock rate reduced to 25Mhz" Added "ADC has systematic gain error when used with input divider in /2, /3 or /4 Mode and when monitoring internal supplies" Added "ADC input Multiplexer is incorrectly documented in data sheet" Added "ADC registers incorrectly documented in data sheet"	5,6,7,8
C	April 19, 2021	Added "Built-in Temperature Monitor (TMON) temperature error may exceed typical error stated in data sheet" errata. Update wording for errata "5.Functional problems with GPIO TX cascade mode"	2,7
D	April 28, 2021	Updated wording for errata "Functional problems with GPIO TX cascade mode"	2

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ERRATA SHEET

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